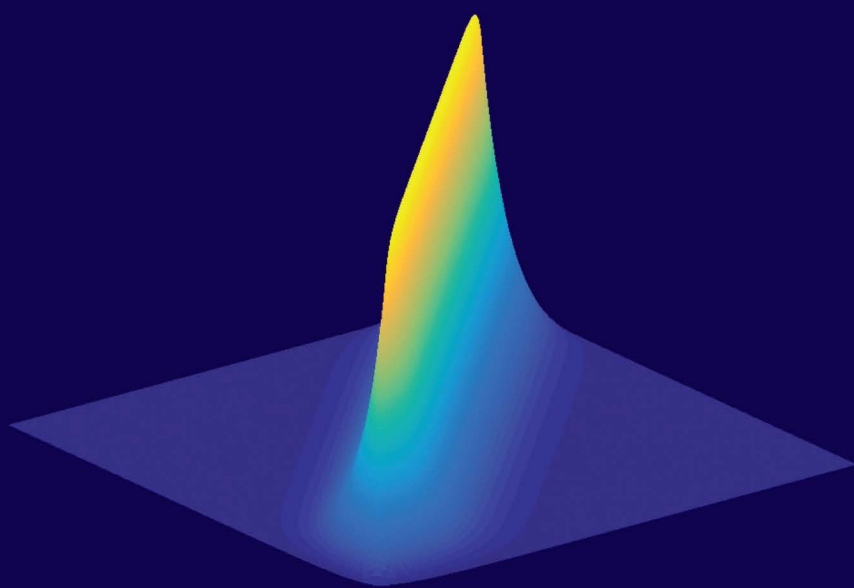


MONOGRAPHS AND RESEARCH NOTES IN MATHEMATICS

Iterative Methods and Preconditioning for Large and Sparse Linear Systems with Applications



Daniele Bertaccini
Fabio Durastante



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Iterative Methods and Preconditioning for Large and Sparse Linear Systems with Applications

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Preface

This book on iterative methods for linear equations can be used as a tutorial and a reference for those who need to solve sparse and/or structured large linear systems of algebraic equations. It can also be used as a textbook for courses in iterative methods or as source material for a course in numerical analysis at the graduate level. It is assumed that the reader is familiar with calculus, elementary numerical analysis, linear algebra, and the main ideas of direct methods for the numerical solution of dense linear systems as described in, e.g., [270]. This text is completed with a bibliography with more than 550 titles where the interested reader can find more details and proofs of results considered here.

The focus is on some real life applications and then on a certain number of methods suitable for them.

This book can be used for a Master's and/or Ph.D. level graduate course in science and engineering and can be useful to researchers, from any field of engineering, health, physical, chemical and biological sciences, etc.

[Chapter 1](#) contains some basic notions of error analysis and sparse matrices. In [Chapter 2](#) there are basic facts on some fundamental iterative algorithms for linear systems, focusing in particular on Krylov subspace methods. [Chapter 3](#) recalls the definition and some general purpose preconditioning strategies. Finally, in [Chapter 4](#), a few examples of preconditioners for Toeplitz and Toeplitz-like, complex symmetric and saddle point systems are considered, together with some introductory notes on multigrid preconditioners.

A short review of linear algebra is also included; see [Appendix A](#).

A number of computational examples and exercises are provided. The former are intended to validate the theoretical results and to give a sense of how the algorithms perform and not to give a complete picture of performance or to be a suite of test problems. Some test matrices and existing *test sets* are mentioned in [Appendix B1](#). Finally, a few numerical linear algebra software packages and repositories are cited in [Appendix B2](#), together with a short description of the codes shipped with the book and how to get them.

D. Bertaccini and F. Durastante



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Symbol Description

A, B, C	Latin upper case letters are used to denote matrices,	$\ \cdot \ _F$	Fröbenius norm, either vectors or matrices,
$\mathbf{u}, \mathbf{v}, \mathbf{w}$	Bold Latin lower case letters are used to denote vectors,	$T_n(f)$	Toeplitz matrix of dimension $n \times n$ generated by the function f ,
$\mathbf{a}_{i,:}, \mathbf{a}_{:,j}$	Denote respectively the i th row, j th column of matrix A ,	$C_n(f)$	Circulant matrix of dimension $n \times n$ generated by the function f ,
α, β, h, k	Greek and Latin lower case letters are used to denote the scalar quantity,	$\mathbb{K}_n$	Space of polynomials of degree less or equal to k over the field $\mathbb{K}$,
$a_{i,j}$	Denote the element of position (i, j) of the matrix A ,	$\langle \cdot, \cdot \rangle_A$	Denotes the scalar product induced by the square matrix A , if A is omitted is the standard scalar product,
$A^{(k)}, \mathbf{u}^{(k)}$	Elements of a sequence of matrices or vectors are denoted by a $(\cdot)$ superscript,	$A \otimes B$	Denotes the Kronecker product between two matrices A and B of opportune size,
α_k, h_k	Elements of a sequence of scalar are denoted by a Latin lower subscript,	$\cdot \circ \cdot$	Denotes the Hadamard product (entrywise product) between two matrices or vectors of opportune size,
$\mathbb{R}, \mathbb{C}, \mathbb{Q}$	Numerical fields, respectively real, complex and rational,	$\cdot^T, \cdot^H$	Denotes respectively the transpose and conjugate transpose operations, either for vectors and matrices,
$\mathbb{N}, \mathbb{Z}$	Sets of the natural and integer numbers, respectively,	$\det(\cdot)$	Determinant of a square matrix,
$\mathcal{C}^k(\Omega)$	Space of the functions of class k over the set Ω ,	$\text{tr}(\cdot)$	Trace of a square matrix,
$\mathcal{C}_0^k(\Omega)$	Space of the functions of class k over the set Ω with compact support,	$\text{rank}(\cdot)$	Rank of a matrix,
$\mathbb{L}^p, \ell^p$	Space of the Lebesgue integrable functions/sequences of order p ,	$\kappa_p(\cdot)$	Condition number in the p norm, if p is omitted $p = 2$,
$\mathcal{B}(U, V)$	space of bounded functions from U to V , if $U = V$ then $\mathcal{B}(U, V) = \mathcal{B}(U)$,	$\text{span}\{\cdot\}$	Linear space generated by some vectors,
$\mathbb{W}^{p,q}$	Sobolev space of order q with p weak derivatives,	$\Lambda(\cdot), \sigma(\cdot)$	Spectrum of a matrix,
$\ \cdot \ _p$	p -norm, either for functions, vectors or matrices ($p = 1, 2, \dots, \infty$),	$\lambda_j(\cdot)$	j th eigenvalue of a matrix,
		$\sigma_j(\cdot)$	j th singular value of matrix.



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Chapter 1

Introduction and Motivations

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The innermost computational kernel of many large-scale scientific applications and industrial numerical simulations, in particular where systems of partial differential equations are involved in the models, is often a sequence of large and sparse linear systems which typically consumes a significant portion of the overall computational time required by the simulation. Many of the matrix problems are in the form of systems of linear equations, although other matrix problems, such as eigenvalue calculations, can occur too. Recent advances in technology have led to a dramatic growth in the size of the matrices to be handled, and iterative techniques are often used in such circumstances, especially when decompositional approaches require prohibitive storage requirements. Computational experience accumulated in the past couple of decades indicates that a good preconditioner holds the key for an effective iterative solver. More recently, the use of parallel computing capabilities of the modern hardware architectures has increased the speed and the range of problems that can be solved; see also § 1.3.

In this book we concentrate on solving linear systems as

$$A \mathbf{x} = \mathbf{b},$$

where for us A will be (if not otherwise stated) a real nonsingular square matrix, $\mathbf{b}$ the known vector and $\mathbf{x}$ the vector of the unknowns.

In order to find an approximation for $\mathbf{x}$ we could use *direct methods*. However, often numerical approximations of many models, in particular of those based on partial differential equations, produce sparse and/or structured matrices A . In general, direct methods are not the best choice because

- they can destroy the underlying sparsity/structure during the resolution process.
- We could be interested in approximations with lower accuracy than the

one provided by, e.g., Gaussian Elimination with pivoting. This could be due to the low precision of the data or to a truncation error introduced by the approximation process generating the linear system(s).

- They cannot use the information given by the initial guess for the solution $\mathbf{x}$. The underlying information is often available (and precious) in many classes of problems.

Both these issues can be fulfilled by *iterative methods*.

In this book we will not consider direct methods. An overview can be found in, e.g., [176, 203, 271, 390].

1.1 Notes on error analysis

Everywhere in this book we refer to absolute and relative error between a vector x and its approximation $\tilde{x}$.

Definition 1.1 (Absolute and relative error). If $\tilde{\mathbf{x}} \in \mathbb{K}^n$ is an *approximation* to $\mathbf{x} \in \mathbb{C}^n$, the *absolute error* in $\tilde{\mathbf{x}}$ relative to the norm $\|\cdot\|$ is defined as $\|\tilde{\mathbf{x}} - \mathbf{x}\|$, while the *relative error* in $\tilde{\mathbf{x}}$ relative to the norm $\|\cdot\|$ is defined as

$$\frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|}{\|\mathbf{x}\|}.$$

All numerical methods proposed here work with the limited precision data that can be handled by our computing machines: *floating point arithmetic* that is an example of *finite precision arithmetic*. At each step, including the simple storage of a floating point number, we can introduce first *rounding errors* that are inherent to the limited precision of the instrument used for our computations. In order to understand this source of errors, assume that, as usual, non-integer numbers are stored in our computers in binary (i.e., in basis 2) with at most t digits. Note that our PCs use a binary *IEEE 754 arithmetics*, see [315], with t up to 53 in Matlab and in the *double precision* arithmetic used by most compilers, including various version of compilers for *Fortran* and *C* languages. In more detail, the *machine numbers* in *IEEE arithmetic* are all and only those real that are 0 or can be written as

$$x = \pm 2^p \sum_{i=1}^t d_i 2^{-i}, \quad d_i = 0 \text{ or } 1, \quad d_1 = 1, \quad -\min \leq p \leq \max, \quad (1.1)$$

where in the *double precision* IEEE 754 arithmetic $\min = -1021$, $\max = 1024$. Of course, not all non-integer numbers can be represented in the format (1.1). Those that cannot must be *rounded* (and therefore approximated and an error introduced) because exact representation can require more than t or even a

not finite number of digits in binary even if in basis 10 they have a finite representation. As an example, 0.1 in basis 10, written as 0.1_{10} for short, in binary is periodic!

$$0.1_{10} = 0.0001100110011 \dots_2.$$

However, an upper bound for the error made storing a non-machine number can be easily computed. Let $fl(x)$ be the machine number used by the computer to store x .

Theorem 1.1. *For all $x \in \mathbb{R}$,*

$$\left| \frac{fl(x) - x}{x} \right| \leq \frac{1}{2} 2^{-t+1}.$$

In order to understand the effects of rounding errors on the accuracy of computations, it is very important to introduce the notion of *Machine precision*.

Definition 1.2 (Machine precision). The number $eps = 2^{-t+1}$ is called here *machine precision*, and eps is the smallest positive machine number such that $fl(1 + eps) > 1$.

For example, *double precision* IEEE 754 arithmetic gives $eps = 2^{-53} \simeq 2.22 \cdot 10^{-16}$.

Usually the error due to the above representation is called *inherent* or *rounding* error. Observe that in general there are at least three other main sources of errors in numerical computations: *data uncertainty*, *truncation* or *analytic* and *algorithmic error*. Data uncertainty and its effects are not investigated here but just recall that often in problems in engineering and economics data is usually accurate to only a few digits and therefore it makes no sense to reduce the other errors much below.

To simplify, the error analysis is based on the assumption that, writing the total error as the sum of the three (four) above mentioned components and neglecting the nonlinear terms, is a reasonable approximation. By calling E_{total} the global error and assuming that the data are exact, we write:

$$E_{total} = E_{inherent} + E_{truncation} + E_{algorithmic} + \delta,$$

where δ includes the nonlinear terms and is considered negligible with respect to the other components.

In the following we suppose $A \in \mathbb{C}^{n \times n}$, $\mathbf{x}, \mathbf{b} \in \mathbb{C}^n$ and that the linear system

$$A\mathbf{x} = \mathbf{b}, \tag{1.2}$$

is consistent. Observe that in a direct method, if no rounding (i.e. representation) errors are present, the solution of the system (1.2) would be computed exactly. On the other hand, for an *iterative method*, even assuming the absence of rounding errors, we need to stop after some iterations, producing a truncation error.

Whatever method we choose, we can not ignore the presence of representation and rounding errors in our computations. It is possible to study these issues by introducing an abstract and precise formulation of the condition of transformations between certain spaces, see e.g. [445]. Here we resume briefly error analysis for problem (1.2). For a more complete treatment of this fundamental issue, see Bevilacqua et al. [81], Higham [304], and in particular the milestone by Wilkinson [542].

1.1.1 Inherent error

The *inherent error*, that *does not depend* on the algorithm selected for the solution of (1.2), measures the sensitivity of the solution to perturbations on the data of the problem, in our case the matrix A and the vector $\mathbf{b}$. Heuristically, the data $A, \mathbf{b}$ are *well-conditioned* if “small” perturbations on them **can** cause “small” changes in the results of the problem (1.2). Otherwise, the data are *ill-conditioned*. In particular, *conditioning* is considered as a characteristic of the **problem**, independently of the algorithm used. A more rigorous meaning of the term *conditioning* can be found in the theorem below.

An analysis of the inherent error can be performed by studying the effect on the solution of perturbations on the data. Following this idea for the case of the linear system in (1.2) gives an important result (see, e.g., [84]).

Theorem 1.2. *Let $\delta A \in \mathbb{C}^{n \times n}$ and $\delta \mathbf{b} \in \mathbb{C}^n$ be the perturbations on the data of problem (1.2) with $\mathbf{b} \neq \mathbf{0}$. Let $\|\cdot\|$ be any induced matrix norm (see appendix A.1). If A is not singular and $\|A^{-1}\|\|\delta A\| < 1$, then also $A + \delta A$ is not singular. If we denote by $\mathbf{x} + \delta \mathbf{x}$ the solution produced on the system by the perturbed data, we have:*

$$(A + \delta A)(\mathbf{x} + \delta \mathbf{x}) = \mathbf{b} + \delta \mathbf{b},$$

where

$$\frac{\|\delta \mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\|A\|\|A^{-1}\|}{1 - \|A^{-1}\|\|\delta A\|} \left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta \mathbf{b}\|}{\|\mathbf{b}\|} \right). \quad (1.3)$$

The scalar quantity $\|A\|\|A^{-1}\|$ is important enough to deserve a definition.

Definition 1.3 (Matrix Condition Number). Given a consistent matrix norm $\|\cdot\|$, the condition number of A is defined as $\kappa(A) = \|A^{-1}\|\|A\|$. Particularly, we denote with $\kappa_p(A) = \|A^{-1}\|_p\|A\|_p$ the matrix condition number with respect to the matrix induced p -norm.

Definition 1.3 permits us to rewrite the bound (1.3) as

$$\varepsilon_{\mathbf{x}} = \frac{\|\delta \mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\kappa(A)}{1 - \kappa(A) \frac{\|\delta A\|}{\|A\|}} \left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta \mathbf{b}\|}{\|\mathbf{b}\|} \right) = \frac{\kappa(A)(\varepsilon_A + \varepsilon_{\mathbf{b}})}{1 - \kappa(A)\varepsilon_A},$$

where $\varepsilon_A = \|\delta A\|/\|A\|$ and $\varepsilon_{\mathbf{b}} = \|\delta \mathbf{b}\|/\|\mathbf{b}\|$ are the perturbations on the

(entries of the) matrix A and on the (entries of the) vector $\mathbf{b}$, respectively. Note that if we just stored A and $\mathbf{b}$ from exact data, then ε_A , $\varepsilon_{\mathbf{b}}$ can be considered of the order of $c \times \text{eps}$, $c \geq 1$ a function of the norm used and here we reasonably assume $c = 1$ to simplify the treatise.

Let us consider few properties of $\kappa(X)$, X a square nonsingular matrix.

1. $\kappa(A) \geq 1$,
2. $\kappa(AB) \leq \kappa(A)\kappa(B)$,
3. $\kappa(\alpha A) = \kappa(A)$ for any $\mathbb{K} \ni \alpha \neq 0$,
4. $\kappa_2(A) = 1$ if and only if A is a nonzero scalar multiple of an orthogonal matrix,
5. $\kappa_2(A) = \kappa_2(A^T)$,
6. $\kappa_2(A^T A) = \kappa_2(A)^2$,
7. $\kappa_2(A) = \sigma_1/\sigma_n$, where σ_1 and σ_n are the largest and smallest singular values of A .

Now, since by property 1 $\kappa(A) \geq 1$, if $\kappa(A)$ takes “small” values then “small” perturbations on the data are expected to give back “small” perturbations on the solution and the problem is said to be *well-conditioned*. On the other hand, if $\kappa(A)$ is “large”, “small” perturbations on the data can give back “large” perturbations on the solution and the problem is *ill-conditioned*. In particular, for simplicity we could consider A well-conditioned if $\text{eps} \cdot \kappa(A) \ll 1$ and ill-conditioned otherwise because of the effects on the relative errors on $\mathbf{x}$, see [Theorem 1.2](#). In general, we should use more care, especially if A is not just a spare matrix but is a term of a sequence of matrices, e.g., $\{A_j\}$, with size increasing with j , say, such as those generated by the discretization of a differential problem; see, e.g., [\[351\]](#), and consider also the behavior of $\kappa(\cdot)$ with respect to j .

Moreover, observe that in a base- b finite precision arithmetic, we can interpret $\log_b(\kappa(A))$ as an estimate of how many base- b digits can be lost in solving our linear system with that matrix independently on the algorithm used, i.e., an estimate of the worst-case loss of precision. Recall that if $\tilde{\mathbf{x}}$ is an approximation of $\mathbf{x}$ with a relative error less than β^{1-t} , we say that $\tilde{\mathbf{x}}$ has t significant digits in the basis β .

Note the differences of *well/ill-conditioned* (problem) with *well/ill-posed* (problem) by Hadamard.

Definition 1.4. A problem is said to be *well-posed* in the Hadamard’s sense if (i) it has a solution, (ii) the solution is unique; (iii) the solution’s behavior changes continuously with the initial conditions. Otherwise, it is *ill-posed*.

Example 1.1 (Hilbert matrices, [\[487, 512\]](#)). A famous class of ill-conditioned

matrices is the Hilbert matrix $H_n(p) \in \mathbb{R}^{n \times n}$ with $p \neq -1, -2, \dots, -(2n-1)$. A Hilbert matrix, for a given value of p , is defined as

$$(H_n(p))_{i,j} = h_{i,j}^{(p)} = (p+i+j-1)^{-1} \text{ for all } i, j = 1, 2, \dots, n.$$

These matrices naturally arise in various situations, e.g., if we want to approximate a continuous function $f(x)$ in the interval $[0, 1]$ by a polynomial of degree $n-1$ in x such that the error

$$E = \int_0^1 (p_{n-1}(x) - f(x))^2 dx, \quad p_{n-1}(x) = \sum_{j=1}^n \alpha_j x^{j-1}$$

is minimized. By differentiating E within respect to the coefficients α_j and imposing the result to be 0 gives its minimum; we obtain the following system of n equation in the α_j s for $i = 1, 2, \dots, n$:

$$\sum_{j=1}^n \left(\int_0^1 x^{i+j-2} dx \right) \alpha_j = \left(\equiv \sum_{j=1}^n h_{i,j}^{(0)} \right) \int_0^1 f(x) x^{i-1} dx \equiv b_i,$$

i.e., $H_n(0)\alpha = \mathbf{b}$. For this particular class of matrices we can express, for nonnegative integer values of p , the inverse explicitly as

$$(H_n^{-1}(p))_{i,j} = \frac{(-1)^{i+j}}{p+i+j-1} \left[\frac{\prod_{k=0}^{n-1} (p+i+k)(p+j+k)}{(i-1)!(n-i)!(j-1)!(n-j)!} \right].$$

In particular, for $n = 4$:

$$H_4(1) = \begin{bmatrix} 1/2 & 1/3 & 1/4 & 1/5 \\ 1/3 & 1/4 & 1/5 & 1/6 \\ 1/4 & 1/5 & 1/6 & 1/7 \\ 1/5 & 1/6 & 1/7 & 1/8 \end{bmatrix},$$

$$H_4^{-1}(1) = \begin{bmatrix} 200 & -1200 & 2100 & -1120 \\ -1200 & 8100 & -15120 & 8400 \\ 2100 & -15120 & 29400 & -16800 \\ -1120 & 8400 & -16800 & 9800 \end{bmatrix}.$$

Therefore we can estimate the row (similarly for the column) sum as

$$\sum_{i=1}^n |(H_n(p))_{i,j}| = \psi(j+n+p) - \psi(j+p),$$

$$\sum_{i=1}^n |(H_n^{-1}(p))_{i,j}| = \frac{\Gamma(n+p+1)\Gamma(j+n+p)}{\Gamma(j)\Gamma(n)\Gamma(-j+n+1)} \cdots$$

$$\cdots \cdots_3 \tilde{F}_2(1-n, j+p, n+p+1; p+1, j+p+1; -1),$$

where $\psi(x)$ is the digamma function and ${}_3\tilde{F}_2(x, y, z)$ is the regularized generalized hypergeometric function, see, e.g., [2]. By using these expressions we can compute the ∞ -norm of both A and A^{-1} and thus compute the condition number for the Hilbert matrix while the $\kappa_2(A)$ can be obtained by looking at the eigenvalues of $H_2(p)$ since it is symmetric, see the values in Table 1.1. Observe what happens when we attempt to solve the interpolation problem

Table 1.1: Example 1.1. Conditions number of Hilbert matrix $H_n(0)$ for various n

n	$\ H_n\ _2$	$\ H_n^{-1}\ _2$	$\kappa_2(H_n)$	$\ H_n\ _\infty$	$\ H_n^{-1}\ _\infty$	$\kappa_\infty(H_n)$
6	1.62	9.24×10^6	1.50×10^7	2.45	1.19×10^7	2.91×10^7
7	1.66	2.86×10^8	4.75×10^8	2.59	3.80×10^8	9.85×10^8
8	1.70	9.00×10^9	1.53×10^{10}	2.72	1.25×10^{10}	3.39×10^{10}
9	1.73	2.86×10^{11}	4.93×10^{11}	2.83	3.89×10^{11}	1.10×10^{12}
10	1.75	9.15×10^{12}	1.60×10^{13}	2.93	1.21×10^{13}	3.54×10^{13}
11	1.77	2.95×10^{14}	5.23×10^{14}	3.02	4.09×10^{14}	1.23×10^{15}
12	1.80	9.54×10^{15}	1.71×10^{16}	3.10	1.33×10^{16}	4.12×10^{16}
13	1.81	3.10×10^{17}	5.63×10^{17}	3.18	4.16×10^{17}	1.32×10^{18}
14	1.83	1.01×10^{19}	1.85×10^{19}	3.25	1.40×10^{19}	4.54×10^{19}
15	1.85	3.31×10^{20}	6.12×10^{20}	3.32	4.64×10^{20}	1.54×10^{21}

for $f(x) = \sin(2\pi x)$ and $n = 15$. To show the effect of the ill-conditioning we perturb only the right-hand side $\mathbf{b}$ with a random vector $\delta\mathbf{b}$ such that $\varepsilon_{\mathbf{b}} \cong 10^{-7}$. The results are reported in Figure 1.1. As expected, observe that

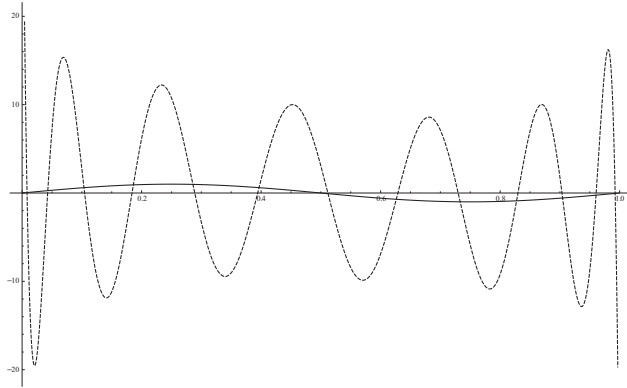


Figure 1.1: Example 1.1. Interpolation problem with Hilbert matrix, comparison between the solution retrieved from the perturbed problem (dashed line) and the objective function (solid line).

$\kappa_2(A) \approx 10^{20}$ and the small perturbations are strongly amplified. We stress again that this effect is independent from the solution algorithm, being only a characteristic of the problem. ✓

1.1.2 Algorithmic error

Let us focus on *algorithmic error* and thus on the algorithm used for computing the actual value $\mathbf{x} = A^{-1}\mathbf{b}$, considering the data exact and studying the rounding errors' propagation. Instead of focusing on the relative error on the quantity $\tilde{\mathbf{x}}$, called *forward* or *a-priori error*, we try to determine which data $\delta\mathbf{b}$ can give $\tilde{\mathbf{x}} = A^{-1}(\mathbf{b} + \delta\mathbf{b})$ instead of $\mathbf{x}$. Clearly, there can be many perturbations $\delta\mathbf{b}$ for which $\tilde{\mathbf{x}}$ can be obtained. We decide to take the perturbation with the smallest norm $\|\delta\mathbf{b}\|$, among the ones for which the equality is obtained. Then, we can consider the ratio $\|\delta\mathbf{b}\|/\|\mathbf{b}\|$ or the value of $\|\delta\mathbf{b}\|$, and call it *backward error*. See Figure 1.2.

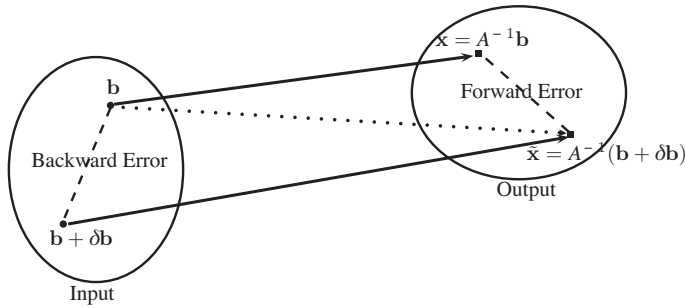


Figure 1.2: Backward and forward error. Continuous line arrows denote the exact computation while dotted line arrows denote the actual computation.

The *backward error analysis* reduces the analysis of *algorithmic error* to finding a bound, as small as possible, to the backward error relative to a computed solution. This goes along with the general observation we made at the end of the previous section about being interested in approximation with lower precision. If the bound from the backward analysis is no larger than the inherent uncertainties, e.g., uncertainties due to other measurements or rounding errors from finite precision arithmetic, then the solution we have obtained can not be distinguished from the solution we are looking for. As a secondary feature, and not less noticeable, *backward error analysis* gives a solid interpretation for the rounding errors as a perturbation in the data, that is a well understood process from Applied Mathematics.

Therefore, it is reasonable to say that an algorithm for computing $\mathbf{x} = A^{-1}\mathbf{b}$ is *backward stable* if, for any $\mathbf{b}$, it produces a computed $\tilde{\mathbf{x}}$ with a backward error that is comparable with the *machine precision*; Definition 1.2.

The relationship between *forward* and *backward error* for a given computational task is governed by the conditioning through the condition number as introduced in Definition 1.3. When all these pieces fit together we can observe that, qualitatively,

$$\text{forward error} \lesssim \text{condition number} \times \text{backward error},$$

with the approximate equality being actually possible.

For many computational tasks, that give reliable results anyway, backward stability can not be proved, therefore a relaxed relation is used. Heuristically we want to accept as *numerically stable* any method that provides almost the right answer for almost the right data. Mixing together backward and forward stability, one usually requires two bounds for the quantities $\delta\mathbf{x}$ and $\delta\mathbf{b}$ in

$$\tilde{\mathbf{x}} + \delta\mathbf{x} = A^{-1}(\mathbf{b} + \delta\mathbf{b}).$$

This weaker formulation is usually known as *mixed forward-backward stability* and whenever the quantities $\delta\mathbf{x}$ and $\delta\mathbf{b}$ can be proven to be “small”, often the algorithm for computing f is said to be *numerically stable*.

Remark 1.1. Even if a numerically stable algorithm is applied to an ill-conditioned problem, like $A\mathbf{x} = \mathbf{b}$ for an ill-conditioned A , it can produce inaccurate results, i.e., even if we bound the backward error, a large condition number can make the forward error very large, that is the *inherent error* becomes dominant. For ill-conditioned problems, even the small bounded errors of stable algorithms can lead to large errors in the approximated solution, consider again [Example 1.1](#).

There exist many well known numerically stable algorithms in Numerical Linear Algebra. Single floating point operations are both forward and backward stable, naïve scalar product algorithms are only backward stable, Gaussian elimination with complete pivoting, triangular backward substitution, QR factorization (both with Housolder and Givens rotations) and the Golub-Kahan algorithm for SVD decomposition are backward stable, see [\[271\]](#) for details. Quite surprisingly, Gaussian elimination with partial pivoting is *not* backward stable, even if the linear equations for which instability is exhibited are extraordinarily rare.

Exercise 1.1. Consider the upper triangular matrices $B_n \in \mathbb{R}^{n \times n}$ of the form,

$$B_n = \begin{bmatrix} 1 & -1 & \dots & -1 \\ 0 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & -1 \\ 0 & \dots & 0 & 1 \end{bmatrix}.$$

Compute $\kappa_\infty(B_n)$, i.e., its condition number in ∞ -norm. Are they ill-conditioned? What happens to the matrix B_n if we set $(B_n)_{n,1} = -2^{2-n}$? Discuss if the determinant of a given matrix A gives any information on its conditioning, does a “small” or a “big” determinant imply anything?

Exercise 1.2. Prove [Theorem 1.1](#).

Exercise 1.3. Prove [Theorem 1.2](#).

1.2 Sparse matrices

Throughout this book we distinguish between *dense* and *sparse* matrices, since in numerical linear algebra the computations made with them needs to be faced in different ways and presents indeed different costs. A first *operative notion* of sparse matrix, independent of the kind of problem, is the one given by Wilkinson in [544]:

“The matrix may be sparse, either with the non-zero elements concentrated on a narrow band centered on the diagonal or alternatively they may be distributed in a less systematic manner. We shall refer to a matrix as dense if the percentage of zero elements or its distribution is such as to make it uneconomic to take advantage of their presence.”

The above is a heuristic starting point that gives the idea: a *sparse* matrix is not *dense*. If we think about a sequence of matrices $A \in \mathbb{R}^{n \times n}$, where n is, e.g., a function of a discretization parameter from a PDE model, we can give a rigorous definition.

Definition 1.5 (Sparse matrix). A matrix A_n of a given sequence of matrices in $\mathbb{R}^{n \times n}$ is sparse if the number of nonzero entries of A , $\text{nnz}(A)$, is $O(n)$.

A useful tool in sparse matrix computation is *graph theory*.

Definition 1.6. ($\text{struct}(A)$) Given a sparse matrix $A \in \mathbb{R}^{n \times n}$, consider the graph $G(A)$, called the structure of A , or $G(A) = \text{struct}(A)$, defined by the vertex set V and edge set E :

$$\begin{aligned} V &= \{i : 1 = 1, \dots, n\}, \\ E &= \{(i, j) : i \neq j \text{ and } A_{i,j} \neq 0\}. \end{aligned}$$

A graphical representation of both $G(A)$ and the sparsity pattern of a matrix A , i.e., a 2D plot in which the axes represents the rows and columns of A and for each nonzero entry of A a point is plotted, is given in [Figure 1.3](#). Observe that in this discussion the *size* of the nonzero entries of the matrix A does not play a role. This can be a limitation, as we will see in the following, see, e.g., § 3.5.1, because in many cases we need to handle matrices that are indeed dense but in which many of the entries are of negligible magnitude. Sometimes these matrices are called *numerically sparse* or *localized* because, by dropping the entries which fall below a threshold magnitude, we can approximate them by sparse matrices; see Benzi in [48] for an interesting overview of localization in matrix computations. Another useful graphic tool is the *city plot*, based on adding a 3rd dimension to the sparsity pattern, either with a color scale or a true zeta axis, on the basis of the magnitude of the elements, see [Figure 1.4](#). Observe also that both *sparse* and *localized* matrices sometimes exhibit a distribution of their entries in some particular patterns.

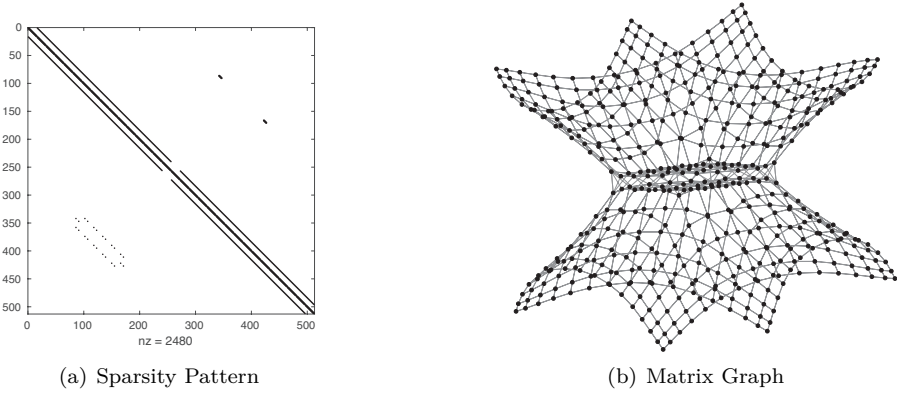


Figure 1.3: Representing sparse matrix: pattern and matrix graph.

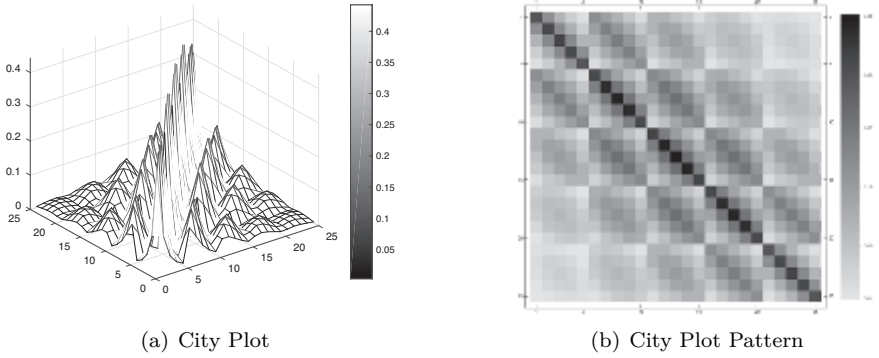


Figure 1.4: Representing sparse matrix: city plot.

We will come back to these topics by considering preconditioners for some *structured matrices* in [Chapter 4](#).

Let us discuss how the presence of many zero entries in the underlying matrices can be used to save computational resources.

1.2.1 Ad-hoc storage and computing schemes for sparse matrices

One of the fundamental basic operations of iterative methods, and usually the most computationally expensive, is the matrix–vector product. Consider the product $A\mathbf{x}$, where A is a generic (and then potentially dense) matrix in $\mathbb{R}^{n \times n}$ and $\mathbf{x}$ a vector in $\mathbb{R}^n$. What is the cost of this operation? We need to perform n scalar products with each row of A and the vector $\mathbf{x}$, $\langle \mathbf{a}_{i,:}, \mathbf{x} \rangle$

for $i = 1, \dots, n$, that have both n elements. Hence, n^2 multiplications and $n(n-1)$ sums. By assuming that a sum and a multiplication take the same computer time to be performed, for short we conventionally call each of them a *flop*, and then we need $2n^2 - n$ flops. On the other hand, if the coefficient matrix A is *sparse*, this computation can be made much more efficiently if the zero entries of A are not stored. To this end, exploiting *sparse storage schemes* and specializing the matrix–vector operations would be the cheapest way to go. In the rest of this section, we use a different paradigm for storing and accessing entries of sparse matrices. In particular, nonzero entries are stored contiguously in an array and other arrays are used for the corresponding indexes. Observe that with this kind of approach, the matrix–vector product algorithm needs to know where the entries are located in the full matrix. Some specialized sparse storage schemes, optimized for the various linear algebra operations, have been investigated, see, e.g., [36, 203, 227, 262, 364, 450, 476, 504]. In the following we use few *storage schemes* that have a well established applicability.

Let us start from the simplest modification of the usual storage scheme, i.e., the *Coordinate Format* (COO). In this format the matrix A is stored into three arrays, e.g., `val`, `row` and `col`. The first is able to accommodate floating point entries of A and the others just the row and column index where the corresponding entry is stored, i.e., `val`(k) has value $a_{i,j}$ and `row`(k) = i and `col`(k) = j , respectively. The matrix–vector product with this format is given by

$$y_i = (A\mathbf{x})_i = \sum_{j=1}^n a_{i,j}x_j \quad \text{for } i = 1, \dots, n, \quad (1.4)$$

as in [Algorithm 1.1](#). The other useful operation we need to investigate is the

Algorithm 1.1: Matrix–vector product with A in COO format

Input: $A \in \mathbb{R}^{n \times n}$ in COO format: `val`, `col`, `row`, $\mathbf{x}$

Output: $\mathbf{y} = A\mathbf{x}$.

```

1  $y_i \leftarrow 0$  for  $i = 1, \dots, n$ ;
2 for  $k = 1, \dots, \text{nnz}(A)$  do
3    $y_{\text{row}(k)} \leftarrow y_{\text{row}(k)} + \text{val}(k)x_{\text{col}(k)}$ ;      /* Scatter and Gather */
```

product $\mathbf{y} = A^T \mathbf{x}$,

$$y_i = (A^T \mathbf{x})_i = \sum_{j=1}^n a_{j,i}x_j, \quad i = 1, 2, \dots, n, \quad (1.5)$$

for a matrix A , stored in COO format, the algorithm is obtained simply by inverting the roles of the arrays `col` and `row` in [Algorithm 1.1](#).

To reduce the number of stored indices, even if they need a relatively

small space because they need just integer arrays, the *Compressed Row Storage* (CRS) and the *Compressed Column Storage* (CCS) are the most popular alternatives. In the CRS format the consecutive nonzero entries of the matrix rows are put in contiguous memory locations. Let us assume that A is sparse and *nonsymmetric*. We need to create 3 arrays: one able to store floating point (or complex) numbers for the nonzero entries of A , **val**, and **col_ind** and **row_ptr**, that contain the (integer) indexes of the entries. Thus, the nonzeros of A are stored in a row-wise fashion in **val**, while the indexes are defined according to the following rule

$$\begin{cases} \text{col_ind}(k) \leftarrow j, & \text{if } \text{val}(k) = a_{i,j}, \\ \text{row_ptr}(i) \leq k < \text{row_ptr}(i+1), & \text{if } \text{val}(k) = a_{i,j}, \\ \text{row_ptr}(n+1) \leftarrow \text{nnz}(A) + 1. \end{cases}$$

In this way, we need to store $2\text{nnz}(A) + n + 1$ integers instead of n^2 . The Compressed Column Storage (CCS), also called the Harwell–Boeing sparse matrix format (see [204]), is defined in an analogue way, except that the columns of A are stored in **val** instead of the rows, i.e., the CCS format for A is the CRS format for A^T . In the case of a symmetric A matrix it is sufficient to store only the upper or the lower triangular part of A .

Example 1.2 (Compress Row/Column Storage). Let us consider the matrix $A \in \mathbb{R}^{4 \times 4}$ given by

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & 3 \\ 4 & 0 & 5 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

Its representation in Compress Row Storage is given by

$$\begin{cases} \text{val} = [1, 2, 1, 3, 4, 5, 1], \\ \text{col_ind} = [1, 2, 1, 4, 1, 3, 2], \\ \text{row_ptr} = [1, 3, 5, 7, 8], \end{cases},$$

while, in Compress Column Storage

$$\begin{cases} \text{val} = [1, 1, 4, 2, 1, 5, 3], \\ \text{row_ind} = [1, 2, 3, 1, 4, 3, 2], \\ \text{col_ptr} = [1, 4, 6, 7, 8], \end{cases},$$

and it easy to verify that the CCS can be obtained as the CRS of A^T . ✓

The matrix–vector product in CRS format is then obtained by means of the usual formula (1.4). Moreover, since we have stored the entries of A by accessing them row-wise, we can implement the matrix–vector multiplication as in Algorithm 1.2. Observe that these methods achieve exactly what we have proposed at the beginning, i.e., we only multiply the nonzero matrix entries. Moreover, the operation count is only $2\text{nnz}(A)$, a saving within respect to the $O(n^2)$ flops of the general *dense* case. This is because performing products

Algorithm 1.2: Matrix–vector product, A stored in CRS format

Input: $A \in \mathbb{R}^{n \times n}$ in CRS format: `val`, `col_ind`, `row_ptr`, $\mathbf{x}$
Output: $\mathbf{y} = A\mathbf{x}$.

```

1 for  $i = 1, \dots, n$  do
2    $y_i \leftarrow 0$ ;
3   for  $j = \text{row\_ptr}(i), \dots, \text{row\_ptr}(i+1) - 1$  do
4      $y_i \leftarrow y_i + \text{val}(j)x_{\text{col\_ind}(j)}$ ;          /* Gather */

```

with zero entries has a nonzero computational cost, that can be non negligible whenever done, e.g., for a hundred thousand times or more.

Unfortunately, we cannot directly use the standard formula (1.5) for computing the product with the transpose matrix. Indeed, this would imply accessing the column of the matrix, an operation that is quite inefficient for matrices stored in the CRS format. Thus, it is better to switch the indices and resort to Algorithm 1.3. The Algorithm 1.3 has the same overall struc-

Algorithm 1.3: Matrix–vector product with A^T in CRS format

Input: $A \in \mathbb{R}^{n \times n}$ in CRS format: `val`, `col_ind`, `row_ptr`, $\mathbf{x}$
Output: $\mathbf{y} = A\mathbf{x}$.

```

1  $y_i \leftarrow 0$  for  $i = 1, \dots, n$ ;
2 for  $j = 1, \dots, n$  do
3   for  $i = \text{row\_ptr}(j), \dots, \text{row\_ptr}(j+1) - 1$  do
4      $y_{\text{col\_ind}(i)} \leftarrow y_{\text{col\_ind}(i)} + \text{val}(i)x_j$ ;

```

ture of Algorithm 1.2 using the same indirect addressing. What is different between the two procedures is the memory access pattern: Algorithm 1.2 at line 1 loads two vectors of data, a row of A and the input vector, and writes out a single scalar. Algorithm 1.3 reads only one element of $\mathbf{x}$ and one row of the matrix A . Both algorithms read and write the vector $\mathbf{y}$. Thus, for A^T , the access to memory limits the overall performance. Moreover, the COO format admits a very fast conversion to and from CSR/CSC formats, thus COO format can be used as a fast way for constructing sparse matrices, while matrix–vector product can then be executed after the CSR/CSC conversion. The following modification of the CRS and CCS storage scheme can be put in place if A is composed by some square dense blocks of nonzeros in some regular pattern. Let us suppose that n_b is the dimension of each block and $\text{nnz}_b(A)$ is the number of nonzero block in $A \in \mathbb{R}^{n \times n}$. In this case the total storage that is needed for the compression is $\text{nnz}_b(A)n_b^2$ and the block dimension n_d of A is the obtained simply by $n_d = n/n_b$. As for the standard CRS scheme, we need 3 arrays for the *Block Compressed Row Storage* (BCRS); these are a rectangular array which access the nonzero block in a

block–row–wise fashion $\text{val}(1, \dots, \text{nnz}_b(A), 1, \dots, n_b, 1, \dots, n_b)$, an integer array $\text{col_ind}(1, \dots, \text{nnz}_b(A))$ which stores the actual column indices of the original matrix A of the first elements of the corresponding nonzero blocks, and a pointer array $\text{row_blk}(1, \dots, n_d + 1)$ whose entries point to the beginning of each block row in the corresponding val and col_ind entry. This kind of storage becomes convenient whenever the matrices have a large n_b . Observe that a *Block Compressed Column Storage* (BCCS) can be considered by storing the sparse nonempty blocks with the CCS format.

If the underlying sparse matrix A is banded, i.e., there exist $p, q \geq 0$ such that $a_{i,j} \neq 0$ if and only if $i - p \leq j \leq i + q$, we can allocate the whole matrix by using the *Compressed Diagonal Storage* (CDS), a rectangular array $n \times (p + q + 1)$ val that permits to store the matrix A by means of the mapping $\text{val}(i, j) \leftarrow a_{i, i+j}$. In the CDS storage scheme, some zeros corresponding to non-existing matrix elements are appended to arrays. In the standard CDS case, the matrix–vector product can be performed either by row or columns, see [Algorithm 1.4](#), while the algorithm for the transpose product is just a minor variation of the latter. The above algorithm accesses the memory by

Algorithm 1.4: Matrix–vector product, A stored in CDS format

Input: $A \in \mathbb{R}^{n \times n}$: val , $p, q \in \mathbb{N}$, $\mathbf{x}$

Output: $\mathbf{y} = A\mathbf{x}$.

```

1  $y_i \leftarrow 0$  for  $i = 1, \dots, n$ ;
2 for  $d = -p, \dots, q$  do
3   for  $l = \max(1, 1 - d), \dots, \min(n, n - d)$  do
4      $y_l \leftarrow y_l + \text{val}(l, d)x_{l+d}$ ;
```

using three vectors per inner iteration. Thus, it can be heavier than the version with the CSR storage. Nevertheless, no indirect addressing is used, and then the algorithm can be easily vectorized. A modification of this format, especially tuned for working with parallel and vector processors is the *Jagged Diagonal Storage* (JDS) format, see [\[387, 429\]](#).

Exercise 1.4. Write an algorithm for computing the product $\mathbf{y} = A^T \mathbf{x}$ for $A \in \mathbb{R}^{n \times n}$ stored in CDS format.

Exercise 1.5. Transform the routine in Algorithms 1.1–1.4 for the matrix–vector product kernels, i.e., $\mathbf{y} = A\mathbf{x}$ and $y = A^T \mathbf{x}$, into kernels for the Basic Linear Algebra Subroutine (BLAS): *SGEMV*, i.e.,

$$\mathbf{y} = \alpha A\mathbf{x} + \beta \mathbf{z}, \quad \text{and} \quad \mathbf{y} = \alpha A^T \mathbf{x} + \beta \mathbf{z},$$

for α, β two scalars and $\mathbf{z} \in \mathbb{R}^n$.

Remark 1.2. Sparse matrices are not the only matrices for which fast matrix–vector product routines can be devised. By means of *displacement formulas* and *fast transformations* it is possible to reduce the computational cost for

matrix–vector operations to less than an $O(n^2)$. Two popular examples are Toeplitz and Circulant matrices, see § 4.1.

Example 1.3 (Matrix vector product). In Figure 1.5 we have the timings for the matrix vector products $\mathbf{y} = A\mathbf{x}$, where

$$A = \begin{bmatrix} T & D_1 & & \\ D_2 & T & \ddots & \\ & \ddots & \ddots & D_1 \\ & & D_2 & T \end{bmatrix}_{n^2 \times n^2}, T = \begin{bmatrix} -4 & 1 & & \\ 1 & -4 & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & -4 \end{bmatrix}_{n \times n},$$

$$D_1 = \text{diag}(-1, \dots, -1), D_2 = \text{diag}(3, \dots, 3),$$

and $\mathbf{x} = (1, \dots, 1)^T$, with A stored in the considered matrix storage formats. We observe that CSR and CSC achieve the best timings. Moreover, the BSR,

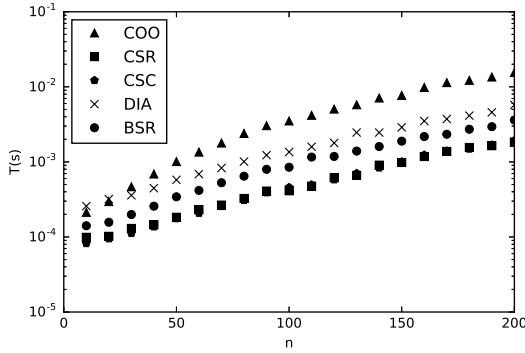


Figure 1.5: Example 1.3. Timings for the matrix–vector product with various sparse storage schemes.

even if A has a very clear block structure, does not beat the CSR/CSC formats since the blocks are itself too sparse to give an effective compression. ✓

1.3 On parallel computing and hardware acceleration

Traditionally we are used to write computer software in a *serial* way, i.e., we construct algorithms as serial streams of instructions in a way that only one instruction can be executed at a time. Nevertheless, many of the core operations that are involved with numerical linear algebra, e.g., matrix–vector products, matrix–matrix products, vector adds etc., can be thought of, and of course implemented, also in a *parallel* form. To accomplish this, one needs to

break the underlying computational task into several independent sub-tasks that can be executed simultaneously. The architectures capable of parallel computing are of various type. We go from a single computer with multiple processors to several computers on the same network to the use of *specialized hardware*. Moreover, any combination of these frameworks is also viable, at least in principle. Why are we interested in moving in this direction? Until the first decade of the twenty-first century, the driving force in satisfying the prediction of Moore's Law¹ [392] has been the *Frequency Scaling* approach: reducing the time for cycle ratio. Nevertheless, this procedure is taxing in two directions: the effective possibility of reducing the size of components and the need of managing power consumption, that depends directly on the processor frequency. Parallel computing and in our specific case, *data parallel*² computing, provides an answer to this problem.

From the point of view of the underlying machinery, here we briefly focus on the approach of *hardware acceleration*, i.e., on the use of a *specialized* piece of hardware that works as an *accelerator*, and *co-processors*. Being more specific, hardware of this kind are the **Graphic Processing Units** (GPUs) and pure processor/co-processor like the Xeon-Phi architectures. Each hardware comes with their own features in term of interfaces, programming languages and way of approaching the computations. Therefore, to make a quick and simple comparison, we look here only at two parameters: the theoretical bandwidth peak and the number of floating point operations per cycle (FLOPS/cycle).

Bandwidth differently from the CPUs, the hardware accelerators are optimized for increasing the **throughput** of the computation, i.e., the number of processes we complete per unit of time. The essential idea is generating a massive number of tasks that can be executed in parallel by each of the computing elements of the accelerator. A wider band gives the ability to perform stream on more data. We measure this quantity in GB/s.

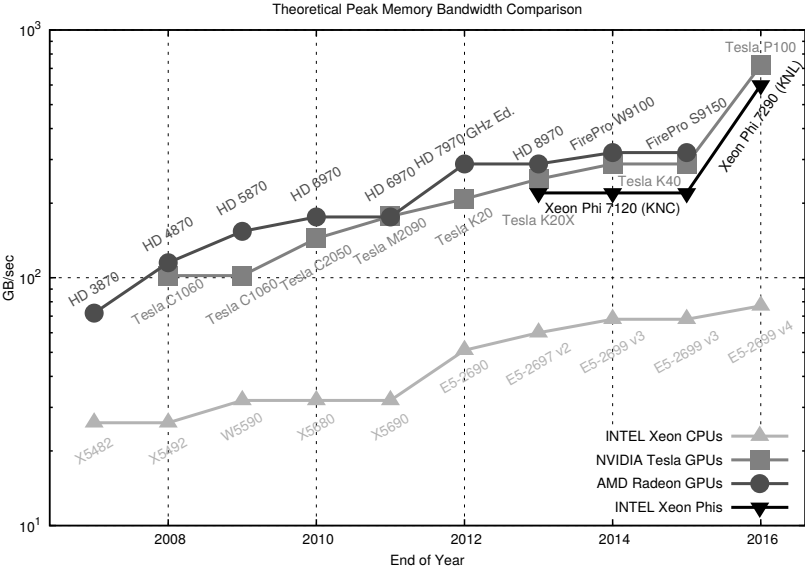
Figure 1.6 reports this data, giving a perspective of the direction in which technology is developing.

Let us now look at a most tangible example of what we have discussed so far.

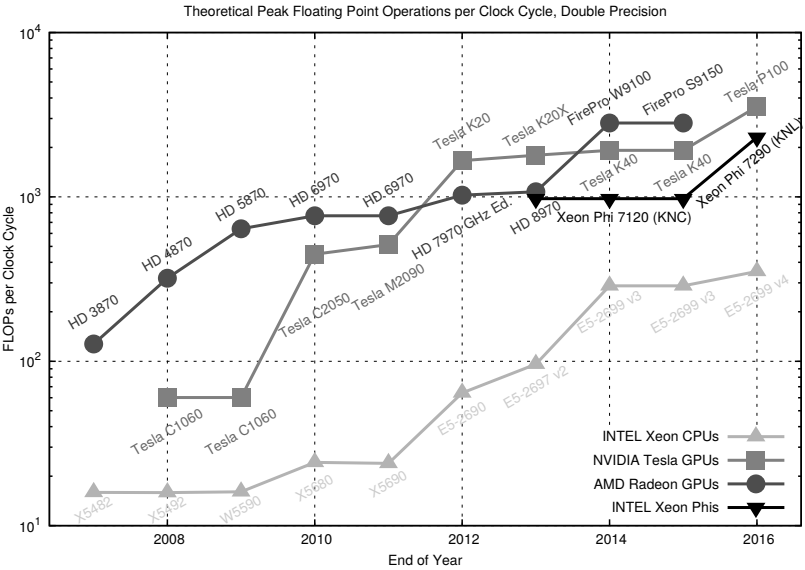
Example 1.4. Consider the core operations for iterative methods, the matrix-vector product $A\mathbf{v}$ with A a sparse matrix. We perform the computation for A of growing size on a CPU, an Intel(R) Xeon(R) CPU E5-2670 v2 @ 2.50GHz, and on a GPU, a NVIDIA GeForce GTX 860M, by using the MATLAB interfaces. As we see from Figure 1.7, the speed-up, measured in term of seconds needed to perform the tasks, obtained with the accelerator is substantial. ✓

¹«The number of transistors in a dense integrated circuit doubles approximately every two years»

²Data parallelism is focused on the distribution of the data, e.g., matrices and vectors, across different computing nodes, which operate on them in parallel.



(a) Bandwidth



(b) FLOPS per Cycle

Figure 1.6: Benchmark over time of some commercial hardware accelerators. Data and elaboration are from www.karlruipp.net.

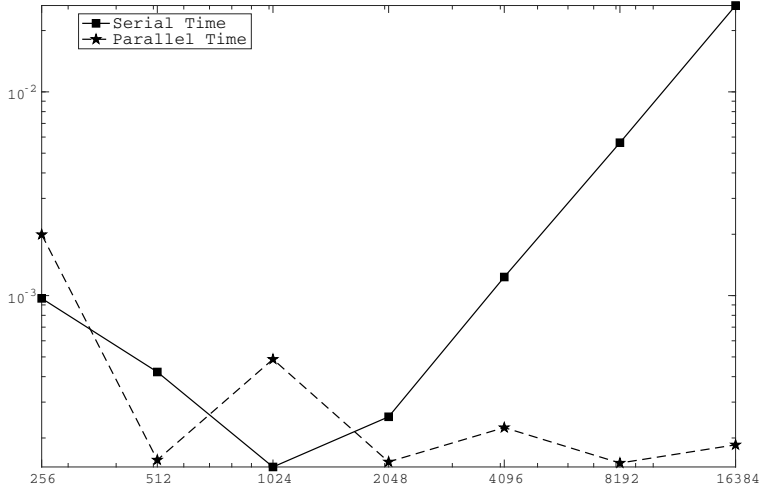


Figure 1.7: Acceleration of the sparse matrix vector product with the GPU in the MATLAB environment. As we see, while the size of the matrix (on the x axis) increases, the GPU accelerated version achieves better timing (on the y axis).

We can not cover the implementation details of the *basic linear algebra subroutines* nor the way to optimize the code and the algorithms for these type of architectures as it would take us out of the target of this book, see, e.g., [39, 120, 175, 366, 545]. Nevertheless, we will try to make it known whenever these strategies can be applied or are available for the material in later chapters.

We conclude this section with some references. For information regarding the news in this area and in large scale computing a good starting point is the website: www.top500.org. For the use of NVIDIA hardware for computations refer to the book by Sanders and Kandrot [463] and the website developer.nvidia.com, while for the use of the AMD hardware and, more in general, the Open Computing Language (openCL) refer to the book by Tay [505] and the websites developer.amd.com and www.khronos.org/opencl. For the INTEL architecture a reference is the book by Jeffers and Reinders [318]. For libraries that implement the BLAS operation in parallel, refer to [172, 200, 230, 516, 517]. Further information is given, when needed, throughout the book.

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